PRD ONE-PAGER · AI IN KYC · DOCUMENT UPLOAD

Route AI-read KYC documents: auto-accept only what a rule can verify

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Proposal: evaluated on specimen data

Problem

On the document-upload path, OCR read PDFs correctly but failed on scans (found in UAT). Reviewing every upload by hand is slow; auto-accepting a wrong PAN is a fraud and compliance risk.

1.00 model confidence on the wrong values: confidence can't be the gate.

Who (v1)

Borrowers who reach the **upload fallback**: DigiLocker, Aadhaar OTP and CKYC didn't work for them. Ops reviewers who check flagged documents.

Out of scope: the DigiLocker, OTP and CKYC routes; video KYC.

Evidence

72 specimen uploads (24 PAN and Aadhaar documents as PDF, scan and photo), exact-match scored. Format checks + confidence ≥ 0.95 auto-accepted **96% (69/72)** with **0** wrong documents let through. Confidence alone let 3 through.

GOAL

Cut manual review on the upload path without letting a wrong PAN or Aadhaar number through, and measure the real error rate instead of assuming it.

REQUIREMENTS

ID	Requirement	Acceptance	Pri.
R1	Extract fields with confidence	Name, DOB, PAN or Aadhaar number, gender, pincode, address returned as structured fields, each with a confidence score.	P0
R2	Format checks before trust	PAN pattern (5 letters, 4 digits, 1 letter), Aadhaar check digit (Verhoeff), valid DOB, 6-digit pincode. Any failure routes to review.	P0
R3	Confidence floor	Auto-accept only if every decision field is ≥ 0.95 and R2 passes. The threshold is a config value, not code.	P0
R4	Address never auto-accepted	No format rule can verify it, so it always goes to review or a separate check.	P0
R5	Review queue	Flagged documents show the image, extracted values and the reason for flagging (which check or field failed).	P0
R6	Random spot-check	A fixed share of auto-accepts goes to a person, so the production error rate is measured.	P0
R7	Read in the background	Extraction runs after upload, not on the live screen (about 18 s per document was observed under load).	P1
R8	Logging	Store raw model output, prompt version, routing decision and reviewer outcome for every document.	P0

SUCCESS METRICS AND PILOT

- **Primary:** share of uploads auto-accepted (test set: 96%).
- **Guardrail:** wrong documents in auto-accepts, from the spot-check (test set: 0 of 69).
- **Also tracked:** review queue time; cost per document (about ₹0.11 at list price).
- **Proof needed:** 0 errors in 69 still allows a true rate up to about 4%. About 300 checked uploads are needed to show it is below 1%.
- **Stop rule:** route everything to review if the spot-check finds a wrong auto-accept, until the cause is fixed.

DEPENDENCIES

- Model API (tested: Gemini 3.5 Flash-Lite) and a data-handling review for sending KYC images to it
- Ops capacity for the review queue and spot-check
- Compliance sign-off on human-in-the-loop for flagged documents

RISKS AND OPEN QUESTIONS

- Real worn cards and budget-phone photos are harder than specimens
- Look-alike characters the format checks can't catch (e.g. inside a name)
- How a larger model or a KYC vendor compares on cost and errors

ROLLOUT

Pilot on the upload path with the spot-check on. After 300+ checked uploads with no wrong auto-accepts, lower the spot-check share. Then add a PAN-to-Aadhaar name and date-of-birth match and an address-to-pincode check.

Evidence: 72 specimen uploads, exact-match scoring against ground truth; specimen data only.

Notebook: Google Colab (linked from the case study)